LF2: THE MEASURE BRIDGE AND BORN-WEIGHT WRAPPER

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ABSTRACT

This paper records the second layer of the Lean4 formalisation programme by isolating the finite-dimensional measure-and-weight layer that sits between deterministic volume typicality and the later operational reconstruction. Building on LF1, which proves that empirical frequencies converge to normalised ontic volume weights under repeated preparation, the present paper formalises the measure bridge from ontic Liouville volume to projective-state weights and then packages the Born-weight layer under explicitly stated external inputs. LF1 itself is not a Born-rule paper, and LF2 is designed to make that separation precise.

The first component is a sector-conditional pushforward result. Under the symmetry-compatible quantum-effective assumptions, with a measurable projection $\pi : \Sigma \to CP^{N-1}$ (smooth and surjective in the underlying physical model), $SU(N)$-equivariance, and invariant Liouville measure μ_L, the pushforward measure is proportional to the Fubini–Study measure:

$$\pi_* \mu_L = c \cdot \mu_{FS}$$

for some constant c.

In the current Lean implementation, the bridge is formalised in this existential form. The explicit mass-comparison formula for c remains part of the paper-level mathematical discussion, but is not yet encoded as a separate Lean theorem.

This is the measure bridge established at the paper level in TN1 and tracked in the programme dependency map as a distinct layer from the later Born-weight step.

The second component is the Born-weight wrapper. Given the projective measure layer, LF2 packages the finite-dimensional probability assignment using an explicit operational consistency package together with an imported Busch effect-Gleason theorem, applied uniformly for finite dimensions N $\geq$ 2.

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1 Introduction

1.1 Role of LF2 within the formalisation programme

The Lean branch of the programme is organised as a layered formalisation of the finite-dimensional CSD stack. LF1 establishes the repeated-trial typicality theorem: under deterministic, measure-preserving ontic dynamics together with an explicit repeated-preparation model, empirical outcome frequencies converge almost surely to the corresponding normalised ontic volume weights. LF1 is therefore the formal statistical backbone of the programme, but it is not yet a Born-rule formalisation. Its result is a theorem about deterministic volume typicality under repeated preparation, not a theorem identifying those weights with the standard projective probability rule. This scope restriction is explicit both in LF1 itself and in the programme dependency map.

LF2 is designed to formalise the next layer. Its task is to connect the ontic volume weights delivered by LF1 to the projective probability structure used in the finite-dimensional reconstruction. In programme terms, LF2 sits between Layer 1, where deterministic dynamics yields stable frequencies, and the later operational layers, where mixed states, POVMs, subsystem reduction, and sequential update are represented in finite dimension. The role of the present paper is therefore narrow but structurally essential: it isolates the measure-and-weight layer as a standalone formal component, with its assumptions and external inputs stated explicitly.

1.2 From deterministic typicality to projective weights

The central conceptual move from LF1 to LF2 is straightforward. LF1 produces normalised ontic weights on the preparation side. If a preparation region $\Omega_0 \subset \Sigma$ and an outcome event are given, then the repeated-trial theorem identifies the almost-sure frequency limit with the corresponding normalised ontic volume ratio. What LF1 does not yet supply is the finite-dimensional projective interpretation of those weights.

That interpretation requires a measure bridge. In the finite-dimensional CSD framework, observational access is modelled in the Lean formal layer by a measurable projection

$$\pi : \Sigma \to CP^{N-1}.$$

In the broader physical model, this projection is also taken to be smooth, surjective, and many-to-one, but only measurability is used in LF2. Once this projection is fixed, the key question becomes measure-theoretic: under what conditions does the pushforward of ontic Liouville measure define the same weight structure as the canonical projective measure on CP^{N-1}? The answer, at the paper level, is supplied by the symmetry-compatible pushforward statement developed in TN1. In the symmetry-compatible quantum-effective sector, if the Liouville measure is invariant and the projection is $SU(N)$-equivariant, then the pushforward measure is proportional to the Fubini–Study measure:

$$\pi_* \mu_L = c \cdot \mu_{FS}$$

for some constant c. This is the first component of LF2.

The second component is operational. Even once μFS has been identified as the canonical projective measure, one still needs the finite-dimensional weight assignment that links projective geometry to the Born form. In the current programme, this step is not treated as a direct consequence of measure uniqueness alone. Rather, it is packaged through an operational consistency layer together with an imported Busch effect-Gleason theorem for effect-additive probability assignments in finite dimensions N $\geq$ 2. Accordingly, LF2 must keep these two layers separate: the measure bridge and the Born-weight wrapper are related, but they are not the same theorem.

1.3 Why LF1 is not yet a Born-rule layer

This distinction matters both mathematically and programmatically. It would be misleading to describe LF1 as a Lean derivation of the Born rule. LF1 formalises the statement that deterministic repeated-preparation models yield stable long-run frequencies equal to normalised ontic weights. It does not formalise the projective measure bridge, does not fix the Fubini–Study measure, and does not prove the finite-dimensional Born-form assignment. Those belong to later layers.

The dependency structure of the programme makes this separation explicit. The first layer is deterministic typicality and frequency emergence. The second layer fixes the admissible projective weight structure in the symmetry-compatible sector. Only after these steps does the programme proceed to the fuller finite-dimensional operational dictionary. LF2 should therefore be read as the formal closure of the measure-and-weight interface, not as a complete formal reconstruction of quantum mechanics.

This point is also methodologically important. One purpose of the Lean branch is to prevent distinct claims from being compressed into a single rhetorical step. In informal foundational writing, deterministic dynamics, invariant measures, projective geometry, and probability assignments are often presented together in a way that obscures which assumptions are doing the work. LF2 is intended to make the logical situation transparent. It does not pretend that all later CSD structure is already formalised. Instead, it records exactly what is needed to pass from LF1's deterministic frequency theorem to the finite-dimensional projective weight layer.

1.4 Main results and scope

The present paper has two principal aims. First, it formalises the symmetry-based measure bridge from ontic Liouville volume to projective Fubini–Study measure. Second, it packages the finite-dimensional Born-weight layer using an explicit operational consistency package together with clearly delimited imported inputs. In this sense, LF2 is a wrapper paper as much as a theorem paper: it identifies the exact formal interface between deterministic ontic weights and the projective probability rule used elsewhere in the finite-dimensional stack.

The scope of LF2 is deliberately limited. The paper works entirely within the finite-dimensional setting. It does not attempt to derive the ontic manifold Σ from deeper principles. It does not derive the projection map π. It does not solve the A5 problem of why physically relevant Hamiltonians lie in the quantum-effective sector. It does not address continuum limits, field-theoretic structure, or the broader scaling questions deferred to later work. Its role is narrower:

to formalise the measure bridge and the Born-weight wrapper that are already logically required by Paper B, TN1, LF1, and the dependency map.

With that scope fixed, the value of LF2 is clear. LF1 shows that deterministic volume typicality yields stable frequencies. LF2 shows how, in the symmetry-compatible finite-dimensional sector, those weights are linked to the canonical projective measure and then to the Born-form probability assignment. The result is a formally cleaner transition from the deterministic statistical backbone to the later finite-dimensional reconstruction layers.

1.5 Status of this document

The present document should be read as a record of the current LF2 implementation state, together with the mathematical role of that implementation within the CSD programme.

2 Ontic and Epistemic Setup

2.1 Ontic space (Σ, μ_L) and finite-dimensional scope

We work throughout in the finite-dimensional setting. The ontic space is taken to be a compact, connected, smooth manifold Σ equipped with a finite Borel measure μ_L. In the intended CSD interpretation, μ_L is the Liouville measure associated with the ontic symplectic structure, but for the logical purposes of the present paper only the measure-theoretic properties required for pushforward and normalisation are used explicitly.

The compactness assumption ensures

$$\mu_L(\Sigma) < \infty,$$

so that all preparation measures considered below are finite and normalisable. This is essential for the passage from ontic weights to projective weights. LF2 does not attempt to derive Σ from deeper principles, nor does it attempt to formalise the full symplectic or Kähler construction from first principles. Those structures belong to the broader finite-dimensional framework and are treated here only through the measure-theoretic data they induce.

The finite-dimensional restriction is not merely technical. The present layer is intended to sit directly above LF1 and directly below the later operational reconstruction papers. Accordingly, it takes as input a finite-dimensional ontic space, a finite Liouville-type measure, and a projection to finite-dimensional projective state space. No infinite-dimensional extension is claimed in this paper.

2.2 Projective epistemic space CP^{N-1}

The epistemic state space is complex projective space

$$CP^{N-1},$$

the space of rays in $\mathbb{C}^N$. This is the canonical operational pure-state space in the finite-dimensional CSD framework. Its geometric structure is fixed independently of any ontic interpretation: it carries the Fubini–Study metric, the associated symplectic form, and the corresponding invariant Borel measure μ_{FS}.

Throughout LF2, μ_{FS} denotes the unnormalised Borel measure associated with the Fubini–Study volume form

$$\Omega_{FS} = \frac{\omega_{FS}^{N-1}}{(N-1)!},$$

matching the convention of TN1. When the normalised version is needed, we write

$$\mu_{FS}^{\text{norm}} := \frac{\mu_{FS}}{\mu_{FS}(CP^{N-1})}.$$

The role of CP^{N-1} in the present paper is twofold. First, it is the codomain of the epistemic projection

$$\pi : \Sigma \to CP^{N-1}.$$

Second, it is the space on which the projective probability weights are defined. LF2 is concerned with the precise relation between these two uses. In particular, the central question is how the ontic measure μ_L induces, via π, a projective measure that can be compared with μ_{FS}.

We reserve uppercase N for the Hilbert-space dimension throughout. This matches the canonical finite-dimensional notation fixed in TN0 and keeps the present formal layer consistent with the wider programme.

2.3 Projection map $\pi : \Sigma \to CP^{N-1}$

The ontic-epistemic link is represented by a measurable map

$$\pi : \Sigma \to CP^{N-1}.$$

In the broader physical model, π is also smooth, surjective, and many-to-one, but only measurability is used in the LF2 formal development. Smoothness and surjectivity are recorded as physical-model assumptions in Appendix A.4 and are not encoded in the Lean module.

The many-to-one character of π is essential. Distinct ontic microstates may correspond to the same epistemic ray, and hence the projective description is coarse relative to the ontic one. This is precisely why the pushforward measure becomes important: a single projective state generally corresponds not to one ontic point, but to an entire fibre

$$F_{[\psi]} := \pi^{-1}([\psi]) \subset \Sigma.$$

LF2 does not require a detailed classification of fibres. It requires only enough regularity of π for the pushforward measure

$$\pi_* \mu_L$$

to be well defined as a finite Borel measure on CP^{N-1}.

Later sections will impose additional structure, specifically symmetry compatibility and $SU(N)$-equivariance, in order to identify this pushforward with the Fubini–Study measure up to scale. At the present stage, however, we record only the basic ontic-epistemic setup.

2.4 Preparation measures, pushforwards, and epistemic densities

Let μ_{prep} denote a preparation measure on Σ. In the standard finite-dimensional setting, μ_{prep} is assumed to be absolutely continuous with respect to μ_L, so that

$$\mu_{\text{prep}} \ll \mu_L.$$

Equivalently, there exists a non-negative density $f_{\text{prep}} \in L^1(\Sigma, \mu_L)$ such that

$$d\mu_{\text{prep}}(x) = f_{\text{prep}}(x)\, d\mu_L(x), \qquad \int_\Sigma f_{\text{prep}}\, d\mu_L = 1.$$

This expresses the preparation as a normalised ontic ensemble relative to the Liouville measure.

Pushing forward along π yields a finite Borel measure on projective space,

$$\pi_* \mu_{\text{prep}},$$

defined in the standard way by

$$(\pi_* \mu_{\text{prep}})(A) := \mu_{\text{prep}}(\pi^{-1}(A))$$

for every Borel subset $A \subseteq CP^{N-1}$. When

$$\pi_* \mu_{\text{prep}} \ll \mu_{FS},$$

one may define the associated epistemic density by the Radon–Nikodym derivative

$$\rho_{\text{ep}} := \frac{d(\pi_* \mu_{\text{prep}})}{d\mu_{FS}},$$

so that

$$\pi_* \mu_{\text{prep}} = \rho_{\text{ep}}\, \mu_{FS}.$$

This is the standard projective density representation used elsewhere in the finite-dimensional stack.

At this stage, two distinct measures must be kept conceptually separate.

First, there is the ontic reference measure μ_L on Σ. Second, there is the canonical projective reference measure μ_{FS} on CP^{N-1}. The purpose of the measure bridge developed in the next section is to connect these two reference structures in the symmetry-compatible sector. Only once that connection is established can one interpret ontic volume weights as projective weights in a canonical way.

Accordingly, the present section provides the minimal input data for LF2: an ontic finite-measure space, a projective epistemic target, a many-to-one projection, and preparation measures whose pushforwards admit comparison with the Fubini–Study structure.

2.5 Symmetry-compatible sector data

The symmetry-compatible sector data consists of the tuple

$$(\Sigma, \mu_L, \pi, \widetilde{U}),$$

where

$$\widetilde{U} : SU(N) \times \Sigma \to \Sigma$$

is a measurable group action on Σ such that:

1. for every $U \in SU(N)$, the map $\widetilde{U}_U : \Sigma \to \Sigma$ is a measurable bijection;
2. μ_L is invariant, in the sense that
$$\mu_L(\widetilde{U}_U(E)) = \mu_L(E)$$
for every Borel set $E \subseteq \Sigma$ and every $U \in SU(N)$;
3. π is equivariant, in the sense that
$$\pi(\widetilde{U}_U(x)) = U \cdot \pi(x)$$
for every $x \in \Sigma$ and every $U \in SU(N)$.

The lifted action $\widetilde{U}$ is part of the input data. It is not constructed from π.

3 Symmetry and the Measure Bridge

3.1 The $SU(N)$ action on CP^{N-1}

The projective space CP^{N-1} carries a natural transitive action of the special unitary group $SU(N)$. For $U \in SU(N)$ and a ray $[\psi] \in CP^{N-1}$, the action is defined by

$$U \cdot [\psi] := [U\psi].$$

This action is well defined on rays and preserves the Fubini–Study metric, the associated symplectic form, and the induced Borel structure. In particular, it defines a symmetry group of the projective state space that is independent of any ontic construction.

The relevance of this symmetry for the present paper is twofold. First, it constrains the class of admissible measures on CP^{N-1}. Second, it provides the compatibility condition under which the ontic pushforward measure may be identified with the canonical projective measure. The measure bridge is therefore not purely measure-theoretic, but symmetry-constrained.

3.2 Uniqueness of the invariant measure on CP^{N-1}

Let μ be a finite Borel measure on CP^{N-1}. We say that μ is $SU(N)$-invariant if

$$\mu(U \cdot A) = \mu(A)$$

for every Borel set $A \subseteq CP^{N-1}$ and every $U \in SU(N)$.

The classification of invariant measures on compact homogeneous spaces implies that, up to normalisation, there is a unique $SU(N)$-invariant Borel probability measure on CP^{N-1}. In the present context, this distinguished measure is the Fubini–Study measure μ_{FS}.

This uniqueness statement is a standard structural input. LF2 does not reprove it in full generality. Instead, it treats the invariant-measure classification as part of the accepted finite-dimensional mathematical background, in the same way that Gleason-class results are treated later in the Born-weight layer.

Combining compactness of CP^{N-1} with the transitive $SU(N)$ action, one obtains the following structural result: any $SU(N)$-invariant Borel probability measure on CP^{N-1} is equal to μ_{FS}.

In particular, if μ is a finite $SU(N)$-invariant Borel measure on CP^{N-1}, then μ is proportional to μ_{FS}. The proportionality constant is determined explicitly in Section 3.3 for the pushforward measure $\pi_*\mu_L$ by comparing total masses on CP^{N-1}.

3.3 Sector-conditional pushforward statement

We now work with the symmetry-compatible sector data of Section 2.5.

Lemma 1 (Preimage-action identity). *Let $A \subseteq CP^{N-1}$ be Borel and let $U \in SU(N)$. Under equivariance of π and bijectivity of $\widetilde{U}_U$,*

$$\pi^{-1}(U \cdot A) = \widetilde{U}_U\big(\pi^{-1}(A)\big).$$

Lemma 2 (Pushforward invariance). *Under Lemma 1 and invariance of μ_L under $\widetilde{U}$, one has*

$$(\pi_*\mu_L)(U \cdot A) = (\pi_*\mu_L)(A)$$

for every Borel set $A \subseteq CP^{N-1}$ and every $U \in SU(N)$.

Theorem 1 (Measure bridge). *Assume the symmetry-compatible sector data of Section 2.5 and the imported invariant-measure uniqueness axiom of Section 7.4. Then*

$$\pi_*\mu_L = c \cdot \mu_{FS}$$

for some constant c.

The value of c is determined by integrating both sides over CP^{N-1}:

$$\mu_L(\Sigma) = c \cdot \mu_{FS}(CP^{N-1}).$$

At the implementation level, the current Lean theorem records the existence of such a constant under the symmetry-compatible sector assumptions and the imported invariant-measure uniqueness axiom. The explicit identification of c by total-mass comparison is not yet stated as a separate Lean theorem.

4 Projective Weights from Pushforward Measure

4.1 Outcome regions on CP^{N-1}

Let M denote a measurement context in the finite-dimensional setting. To each such context we associate a finite collection of measurable outcome regions

$$\{\Omega_i^{\mathrm{ep}}(M)\}_{i \in I},$$

where each $\Omega_i^{\mathrm{ep}}(M) \subseteq CP^{N-1}$ is a Borel subset of projective space, and the family forms a partition up to sets of μ_{FS}-measure zero:

$$\mu_{FS}\left(CP^{N-1} \setminus \bigcup_{i \in I} \Omega_i^{\mathrm{ep}}(M)\right) = 0, \qquad \mu_{FS}\big(\Omega_i^{\mathrm{ep}}(M) \cap \Omega_j^{\mathrm{ep}}(M)\big) = 0 \text{ for } i \neq j.$$

These regions represent the epistemic realisation of measurement outcomes. Their construction is context-dependent, and LF2 does not attempt a full classification. It requires only that they are measurable and that they form a partition in the above sense.

For the Lean module, the index set I is taken to be $\mathrm{Fin}\, n$ for some $n \in \mathbb{N}$, and the partition is encoded as a `MeasurablePartition` structure with fields:

$$\begin{aligned}
&\texttt{parts} : \mathrm{Fin}\, n \to \mathrm{Set}(CP^{N-1}),\\
&\texttt{measurable} : \forall i,\ \mathrm{MeasurableSet}(\texttt{parts}\ i),\\
&\texttt{pairwise_null} : \forall i\, j,\ i \neq j \to \mu_{FS}\big((\texttt{parts}\ i) \cap (\texttt{parts}\ j)\big) = 0,\\
&\texttt{cover_null} : \mu_{FS}\left(CP^{N-1} \setminus \bigcup_i \texttt{parts}\ i\right) = 0.
\end{aligned}$$

4.2 Weights induced by the prepared pushforward measure

Given the measure bridge of Theorem 1,

$$\pi_*\mu_L = c \cdot \mu_{FS}$$

for some constant c, one obtains a natural assignment of weights to outcome regions.

Let μ_{prep} be a preparation measure on Σ, absolutely continuous with respect to μ_L, and let

$$\pi_*\mu_{\text{prep}} = \rho_{\text{ep}}\, \mu_{FS}$$

be the corresponding epistemic measure on CP^{N-1}. For each outcome region $\Omega_i^{\text{ep}}(M)$, define the weight

$$w_i := (\pi_*\mu_{\text{prep}})\big(\Omega_i^{\text{ep}}(M)\big) = \int_{\Omega_i^{\text{ep}}(M)} \rho_{\text{ep}}\, d\mu_{FS}.$$

Equivalently, in purely ontic terms,

$$w_i = \mu_{\text{prep}}\big(\pi^{-1}(\Omega_i^{\text{ep}}(M))\big)\,.$$

This expresses the projective weight as the ontic measure of the corresponding pulled-back outcome set.

4.3 Normalisation and state dependence

By construction, the weights satisfy

$$\sum_{i\in I} w_i = 1,$$

since the outcome regions form a partition up to μ_{FS}-null sets and $\pi_*\mu_{\text{prep}}$ is a probability measure.

The weights depend on the preparation through the density ρ_{ep}. In particular, for a sharply prepared epistemic state concentrated near a ray $[\psi] \in CP^{N-1}$, the measure $\pi_*\mu_{\text{prep}}$ is concentrated near $[\psi]$, and the weights w_i reflect the relative overlap of this concentration with the outcome regions.

At this stage, LF2 does not assume a specific functional form for ρ_{ep}. The dependence on $[\psi]$ is encoded entirely through the preparation measure and its pushforward. The next section will introduce the additional structure required to identify this dependence with the standard Born-form assignment.

4.4 Relation to prepared epistemic densities

The expression

$$w_i = \int_{\Omega_i^{\text{ep}}(M)} \rho_{\text{ep}}\, d\mu_{FS}$$

is the general finite-dimensional probability formula at the projective level. It should be contrasted with state-independent expressions of the form

$$\frac{\mu_{FS}(\Omega_i^{\text{ep}})}{\mu_{FS}(CP^{N-1})},$$

which are not, in general, physically meaningful unless ρ_{ep} is uniform.

Thus, the role of the measure bridge is not merely to identify μ_{FS} as a canonical measure, but to provide the reference structure relative to which epistemic densities are defined. The actual probability weights are always preparation-dependent, and this dependence is encoded in ρ_{ep}.

This completes the construction of projective weights from the pushforward measure. The next section packages the additional operational input required to recover the Born-form assignment in finite dimension.

5 Born-Weight Wrapper

5.1 Operational consistency package

Definition 5.1 (Operational consistency package)**.** Let H be a finite-dimensional complex Hilbert space. An operational consistency package consists of a map

$$p : \mathcal{E}(H) \to [0, 1]$$

from the effect algebra $\mathcal{E}(H)$ to the unit interval such that:

1. $p(E) \geq 0$ for every $E \in \mathcal{E}(H)$;
2. $p(E) \leq 1$ for every $E \in \mathcal{E}(H)$;
3. $p(I) = 1$;

4. whenever $E, F \in \mathcal{E}(H)$ satisfy $E + F \leq I$, one has
$$p(E + F) = p(E) + p(F).$$

The current Lean implementation does not include the unitary-covariance clause as a field of the core `OperationalPackage` structure. The reason is formal rather than conceptual. The naive invariance reading over-constrains the package, while the genuinely covariant reading is type-heavier and is deferred to LF3. The present repository instead implements the structural helper `Effect.conjugateBy` and leaves the full covariance packaging to the next layer.

5.2 Imported effect-Gleason theorem

In the implemented Lean development, the relevant imported theorem is the axiom `busch_effect_gleason`. It is stated for an `OperationalPackage N`, a dimension hypothesis $2 \leq N$, and a resulting unique density operator ρ : `DensityOperator N` satisfying the trace-form representation on all effects.

Both the invariant-measure uniqueness theorem and the Busch effect-Gleason theorem are stated as `axiom` declarations rather than imported from Mathlib, because as of this writing neither theorem is present in Mathlib.

5.3 Uniform treatment of the qubit case

Under the effect-additive package of Definition 5.1, the qubit case requires no separate treatment.

5.4 Finite-dimensional Born-form assignment

Combining the measure bridge with the operational consistency package of Definition 5.1 and the imported Busch effect-Gleason theorem yields the finite-dimensional Born-form assignment.

For a pure preparation $[\psi] \in CP^{N-1}$ and a measurement with projectors $\{P_i\}$, the weights take the form
$$w_i = \langle\psi|P_i|\psi\rangle.$$

Equivalently, writing $P_i = |\phi_i\rangle\langle\phi_i|$ for rank-one projectors,
$$w_i = |\langle\psi|\phi_i\rangle|^2.$$

This is the standard quadratic probability rule. In the present framework, it is not attributed to a single theorem. Rather, it arises from the combination of:

- the measure bridge, which identifies the canonical projective measure;
- the preparation-dependent density ρ_{ep} on CP^{N-1};
- the operational consistency package;
- the imported Busch effect-Gleason theorem.

The role of LF2 is to formalise this interface. It makes explicit how the deterministic volume weights from LF1 are connected, through symmetry and operational structure, to the Born-form assignment used in the finite-dimensional reconstruction.

At the implementation level, the current repository proves a genuine quadratic pure-state identity inside Lean. In particular, the theorem `born_quadratic` establishes
$$\mathrm{Tr}(|\psi\rangle\langle\psi|\,|\phi\rangle\langle\phi|) = \|\langle\psi, \phi\rangle\|^2$$
for unit vectors ψ, ϕ. A strengthened endpoint theorem, `pure_state_born_weights_of_certainty`, then derives the rank-one Born weights from the imported Busch effect-Gleason theorem together with a pure-state certainty hypothesis.

6 Interface with LF1

6.1 LF1 output as normalised ontic weight

LF1 establishes the repeated-trial typicality theorem for deterministic ontic dynamics under a preparation model. For a given preparation region $\Omega_0 \subset \Sigma$ and an outcome event represented by a measurable set $\Omega_i^{\Sigma} \subset \Sigma$, the LF1 result states that the empirical frequency converges almost surely to the corresponding normalised ontic weight:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \mathbf{1}_{\Omega_i^\Sigma}(x_k) = \frac{\mu_L(\Omega_i^\Sigma \cap \Omega_0)}{\mu_L(\Omega_0)}.$$

Equivalently, writing μ_{prep} for the conditional preparation measure on Ω_0, this can be expressed as

$$w_i^{\mathrm{ontic}} = \mu_{\mathrm{prep}}(\Omega_i^\Sigma).$$

This is the output of LF1: a deterministic construction of stable weights defined entirely at the ontic level.

6.2 From frequency convergence to projective weights

LF2 does not modify the LF1 result. Instead, it reinterprets the ontic weight in projective terms.

Given a measurement context M, let $\Omega_i^{\mathrm{ep}}(M) \subset CP^{N-1}$ be the corresponding epistemic outcome region. The associated ontic outcome set is then given by the pullback

$$\Omega_i^\Sigma(M) := \pi^{-1}(\Omega_i^{\mathrm{ep}}(M)).$$

The LF1 weight can therefore be written as

$$w_i^{\mathrm{ontic}} = \mu_{\mathrm{prep}}\big(\pi^{-1}(\Omega_i^{\mathrm{ep}}(M))\big).$$

By definition of the pushforward measure, this is exactly

$$w_i^{\mathrm{ontic}} = (\pi_*\mu_{\mathrm{prep}})\big(\Omega_i^{\mathrm{ep}}(M)\big).$$

Thus the LF1 weight coincides with the projective weight defined in Section 4:

$$w_i = w_i^{\mathrm{ontic}}.$$

This identity is purely measure-theoretic. It does not yet invoke any symmetry or Born-form structure. It is simply the statement that the LF1 weights, when expressed through the projection map, are exactly the pushforward weights on CP^{N-1}.

The Lean-side encoding of this identity is specified in Section 7.6.

6.3 Separation of theorem layers

The purpose of the present section is to make the logical separation between LF1 and LF2 explicit.

- LF1 proves convergence of empirical frequencies to ontic weights:
$$\text{frequency} \to \mu_{\mathrm{prep}}(\Omega_i^\Sigma).$$
- The projection map π converts ontic outcome sets into epistemic regions:
$$\Omega_i^\Sigma = \pi^{-1}(\Omega_i^{\mathrm{ep}}).$$
- The pushforward measure identifies these weights with projective weights:
$$\mu_{\mathrm{prep}}(\Omega_i^\Sigma) = (\pi_*\mu_{\mathrm{prep}})(\Omega_i^{\mathrm{ep}}).$$
- The measure bridge and Born-weight wrapper then determine the functional form of these weights.

Each step uses a different input:

- deterministic dynamics and repeated preparation (LF1),
- projection structure,
- symmetry and invariant measure,
- operational consistency and the imported Busch effect-Gleason theorem.

LF2 does not collapse these into a single argument. It records them as distinct layers.

6.4 What LF2 adds beyond LF1

LF1 establishes that deterministic volume typicality yields stable frequencies equal to ontic weights. LF2 adds two further pieces of structure:

1. the measure bridge, which identifies the canonical projective measure induced by the ontic measure;
2. the Born-weight wrapper, which fixes the admissible probability assignment on CP^{N-1} under operational constraints.

Together, these show that the LF1 weights are not only stable, but coincide with the standard finite-dimensional probability rule when expressed in projective terms.

Importantly, LF2 does not extend beyond this interface. It does not derive the projection map, does not resolve the quantum-effective sector, and does not address composite or continuum structure. Its contribution is to make the transition from deterministic ontic weights to projective Born-form weights precise and formally separable from the surrounding assumptions.

7 Lean4 Formalisation Strategy

7.1 Abstract measure layer

The formalisation of LF2 is designed to sit directly on top of the LF1 infrastructure. Accordingly, the starting point is the abstract measure-theoretic layer already implemented in LF1, where the ontic space is represented as a measurable type Σ equipped with a finite measure μ_L.

LF2 extends this layer by introducing a second measurable space P, standing for the abstract projective target, together with a measurable projection

$$\pi : \Sigma \to P.$$

In the implemented repository, this data is bundled into a structure `SectorData` consisting of an LF1 ontic setup, a measurable projection, an ontic action $G \to \Sigma \simeq^m \Sigma$, an epistemic action $G \to P \simeq^m P$, coherence fields for the group law, Liouville-measure invariance of the ontic action, and equivariance of the projection.

At the Lean level, the symmetry-compatible sector should be represented by a structure carrying:

- a measurable type Σ,
- a finite measure μ_L,
- a measurable projection $\pi : \Sigma \to CP^{N-1}$,
- a measurable $SU(N)$ action on Σ, for example as

  $$\texttt{action} : \texttt{SU}\,\texttt{N} \to \texttt{MeasurableEquiv}\,\Sigma\,\Sigma$$

  or an equivalent measurable-action formulation,
- proof fields expressing equivariance of π and invariance of μ_L.

7.2 Encoding projective-space structure

While CP^{N-1} carries rich geometric structure, the Lean formalisation focuses on the minimal data required for the measure bridge. In particular, the projective space is treated as a measurable space equipped with a distinguished reference measure μ_{FS}.

The action of $SU(N)$ is not constructed in full differential-geometric detail. Instead, the formalisation encodes the invariance properties that are required for the uniqueness of the measure. This allows the argument to proceed without committing to a specific coordinate realisation of CP^{N-1}.

7.3 Pushforward and invariance statements

The central formal object is the pushforward measure

$$\pi_* \mu_L,$$

implemented in Lean using the standard pushforward construction for measures.

The uniqueness of the invariant measure is represented by an explicit imported axiom, for example

`axiom invariant_measure_uniqueness_CPN.`

In the formalisation, the equivariance and invariance conditions are introduced as explicit hypotheses. The uniqueness of the invariant measure is treated as an external input, represented either as an axiom or as a previously established theorem in the formal library.

In the current implementation, the core theorem chain is split into the named results `SectorData.pushforward_apply`, `SectorData.preimage_action_eq`, `SectorData.pushforward_epAction_invariant`, and `measure_bridge`. This decomposition mirrors the theorem structure of the manuscript while matching the actual modular proof layout in the repository.

7.4 Imported external theorem boundary

A key design principle of LF2 is to make the boundary between internal formalisation and external mathematical input explicit.

Three named components are currently treated as imported:

- the uniqueness of the $SU(N)$-invariant measure on the abstract projective target;
- the Busch effect-Gleason theorem used in the Born-weight wrapper;
- a rank-one density uniqueness lemma stating that a density operator assigning certainty to a rank-one projector must itself be that rank-one projector.

In Lean, these appear as explicit named axioms rather than as internally reproved theorems. This makes the external theorem boundary fully transparent. The present LF2 implementation therefore machine-verifies the wrapper layer relative to these imported inputs, rather than claiming a fully self-contained derivation from first principles inside Lean.

7.5 Repository architecture and dependency flow

The current repository is no longer merely prospective. LF2 is exported from the top-level file `CsdLean4.lean`, which imports the following LF2 modules:

- `CsdLean4/LF2/Setup.lean`: sector data over the LF1 ontic setup;
- `CsdLean4/LF2/MeasureBridge.lean`: pushforward identities, invariance transfer, and the measure-bridge theorem;
- `CsdLean4/LF2/Weights.lean`: measurable partitions and projective weights;
- `CsdLean4/LF2/BornWrapper.lean`: effect and density-operator structures, the Busch axiom, the quadratic Born identity, and pure-state Born-weight theorems;
- `CsdLean4/LF2/Interface.lean`: the LF1 to LF2 weight identity and the combined LF1 plus LF2 theorem.

The dependency flow is strictly layered:

- LF1 provides the repeated-trial convergence theorem and ontic weights;
- LF2 `Setup` defines the abstract sector data;
- `MeasureBridge` establishes pushforward invariance and the bridge theorem;
- `Weights` defines projective weights and partition normalisation;
- `BornWrapper` packages the finite-dimensional Born-weight layer;
- `Interface` links the LF1 output to the LF2 projective-weight structure.

This modular structure mirrors the conceptual layering of the programme. It ensures that each component can be verified independently and that later extensions, such as mixed states and sequential measurement, can build on a clearly defined base.

Note: The MeasurablePartition structure is defined in Weights.lean.

7.6 LF1 interface in Lean

LF2 imports the implemented LF1 backbone through the module chain `Setup.lean`, `Preparation.lean`, `Outcomes.lean`, `Trials.lean`, `Indicators.lean`, `Expectation.lean`, `Convergence.lean`, and `MainTheorem.lean`.

The implemented LF1 interface consists of the theorems `lf1_weight_eq_projective_weight`, `lf1_prepMeasure_weight_eq_projective_weight`, and `LF1_main_theorem_projective`. These identify the LF1 ontic weight with the LF2 projective weight under pushforward and then lift the LF1 frequency theorem into projective-weight language.

The confirmed LF1 names required by LF2 are:

- `LF1_main_theorem_ae`: the exported almost-everywhere frequency-convergence theorem;
- `prepMeasure_apply`: the explicit rewriting lemma for the preparation measure;
- `TrialModel` and `OutcomeRegion`: the LF1-side repeated-trial and outcome-region objects.

If the repository confirms additional structure names such as `OnticSetup`, those exact implementation names should be used here. The present document should follow the repository naming rather than introducing speculative aliases.

The LF2-side interface theorem should be stated as:

$$\mu_{\text{prep}}\big(\pi^{-1}(\Omega_i^{ep}(M))\big) = (\pi_*\mu_{\text{prep}})\big(\Omega_i^{ep}(M)\big),$$

with the Lean-side interface theorem named, for example, `lf1_weight_eq_projective_weight`.

7.7 Mathlib dependencies

The intended Mathlib dependencies are as follows.

- `MeasureTheory.Measure.Basic` for `Measure`, `map`, and pushforward constructions;
- `MeasureTheory.Measure.MeasureSpace` for measurable-space structure;
- `MeasureTheory.MeasurableSpace.Basic` for measurability hypotheses;
- `Mathlib.Algebra.Group.Defs` for `MulAction`, together with explicit measurability hypotheses on the action.

No Jacobian or change-of-variables API is needed at this stage.

The pushforward identity

$$(\pi_*\mu_L)(A) = \mu_L(\pi^{-1}(A))$$

is implemented through `MeasureTheory.Measure.map_apply`, requiring `Measurable` π and `MeasurableSet` A.

The invariant-measure uniqueness theorem on CP^{N-1} is not currently available in Mathlib and should therefore be introduced as an imported axiom, for example `invariant_measure_uniqueness_CPN`.

8 Scope and Limitations

8.1 Finite-dimensional restriction

All results in this paper are stated in the finite-dimensional setting. The ontic space Σ, the projection π, and the epistemic space CP^{N-1} are all assumed to be finite-dimensional. The measure bridge and the Born-weight wrapper are formulated entirely within this regime.

No claim is made that the present construction extends directly to infinite-dimensional Hilbert spaces or to continuum systems. The purpose of LF2 is to formalise the finite-dimensional layer that sits immediately above LF1. Any extension beyond this setting requires additional structure and is deferred to later work.

8.2 Dependence on symmetry-sector assumptions

The measure bridge is not a general theorem for arbitrary ontic projections. It is a sector-conditional result. In particular, it depends on symmetry-compatibility assumptions, including:

- invariance of the ontic measure μ_L under a lifted symmetry action;
- $SU(N)$-equivariance of the projection map π.

These conditions define the symmetry-compatible quantum-effective sector in which the pushforward measure can be identified with the Fubini–Study measure up to scale. LF2 does not derive these assumptions. It makes them explicit and uses them as hypotheses.

8.3 No derivation of Σ or π

The ontic manifold Σ and the projection map

$$\pi : \Sigma \to CP^{N-1}$$

are treated as given. LF2 does not attempt to derive their existence, uniqueness, or detailed structure from deeper principles.

In particular, the geometry of Σ, the origin of its symplectic or Kähler structure, and the physical interpretation of the projection map are outside the scope of the present paper. LF2 uses only the measure-theoretic and mapping properties required to define pushforward measures and projective weights.

8.4 External inputs in the Born-weight layer

The Born-weight wrapper relies on external mathematical inputs. These currently include:

- the uniqueness of the $SU(N)$-invariant measure on the abstract projective target;
- the Busch effect-Gleason theorem for finite-dimensional effect-additive operational packages;
- a rank-one density uniqueness lemma used to tighten the pure-state endpoint from a certainty hypothesis to the standard quadratic Born form.

LF2 does not rederive these results. It incorporates them as explicit assumptions within the formalisation. The role of the present paper is to show how these inputs connect to the measure bridge and to the LF1 output, not to replace them.

8.5 Separation from later reconstruction layers

LF2 does not attempt to provide a full reconstruction of finite-dimensional quantum mechanics. In particular, it does not address:

- mixed states and convex structure;
- general POVM constructions beyond the imported results;
- subsystem reduction and composite structure;
- sequential measurement and update rules.

These topics belong to later layers of the programme and are treated in subsequent notes. The purpose of LF2 is limited to the measure-and-weight interface.

8.6 No claim beyond the weight layer

The results of LF2 should be read as establishing a specific interface:

- LF1 provides deterministic frequency convergence to ontic weights;
- LF2 identifies the corresponding projective measure structure;
- LF2 packages the Born-form assignment under explicit assumptions.

No broader claim is made. In particular, LF2 does not claim:

- a derivation of the quantum-effective sector;
- a classification of admissible Hamiltonians;
- an extension to continuum or field-theoretic regimes.

These remain open problems in the programme. The present paper isolates the measure-and-weight layer so that these later questions can be addressed on a clearly defined foundation.

9 Conclusion

9.1 Summary of results

This paper has formalised the measure-and-weight layer that connects deterministic ontic dynamics to the finite-dimensional projective probability structure.

The first result is the measure bridge. Under symmetry-compatible assumptions, the pushforward of the ontic Liouville measure defines a projective measure proportional to the Fubini–Study measure on CP^{N-1}:

$$\pi_* \mu_L = c \cdot \mu_{FS},$$

for some constant c.

The second result is the Born-weight wrapper. Given the projective measure structure, and under the operational consistency package of Definition 5.1 together with the imported Busch effect-Gleason theorem, the admissible probability assignment takes the standard quadratic form for finite-dimensional systems.

These two components are logically distinct. The measure bridge identifies the canonical projective measure. The Born-weight wrapper fixes the admissible probability assignment relative to that measure. LF2 makes this separation explicit and formal.

9.2 Role within the formalisation stack

LF2 completes the second layer of the Lean formalisation programme.

LF1 establishes deterministic frequency convergence to ontic weights. LF2 identifies the corresponding projective measure structure and packages the finite-dimensional probability rule. Together, these two layers provide a complete interface between deterministic volume typicality and the projective probability structure used in the later reconstruction.

This layered structure is intentional. It ensures that the statistical, geometric, and operational components of the framework are not conflated. Each layer is stated with its own assumptions and its own conclusions.

9.3 What is established

Within its stated scope, LF2 establishes the following:

- a symmetry-constrained identification of the pushforward measure with the Fubini–Study measure;
- a precise definition of projective weights induced by ontic preparation measures;
- a formal interface between deterministic frequency convergence and projective probability assignment;
- an explicit separation between internally formalised results and externally imported inputs.

These results complete the finite-dimensional measure-and-weight layer required by the earlier papers.

9.4 What remains open

LF2 does not resolve several central questions in the programme.

In particular, it does not:

- derive the ontic manifold Σ or the projection map π;
- explain the origin of the symmetry-compatible sector;
- address composite-system structure from first principles;
- extend the construction beyond finite dimension.

These are not omissions. They are part of the staged structure of the programme. LF2 isolates the layer that can be formalised independently of these deeper questions.

9.5 Next steps

With the measure-and-weight layer in place, the next steps in the formalisation programme are clear.

Subsequent work will extend the formalisation to:

- mixed states and convex structure;
- general measurement frameworks, including POVMs;
- subsystem reduction and composite systems;
- sequential measurement and update rules.

These layers build directly on the interface established by LF1 and LF2.

9.6 Final remark

The primary contribution of LF2 is not a single new theorem, but a clarification of structure. It identifies the exact point at which deterministic ontic weights become projective probability weights, and it records the assumptions required for that transition.

In doing so, it separates what is proved, what is assumed, and what is deferred. That separation is essential for any further development of the formalisation programme.

A Assumptions Audit

A.1 Purpose of the audit

A central aim of LF2 is not only to formalise the measure bridge and Born-weight wrapper, but also to make their logical dependencies completely explicit. LF1 already adopted this approach by separating internally proved statements, imported mathematical results, and physical-model assumptions, precisely because formalisation forces the underlying theorem architecture to be stated without compression. The same discipline is even more important here, since LF2 sits at the point where deterministic ontic weights are connected to projective probability structure through a combination of internal formalisation and external theorem inputs.

For that reason, the present section separates the ingredients of LF2 into four distinct classes:

1. statements proved internally in the formal development;
2. mathematical results imported as external inputs;
3. assumptions that belong to the physical model rather than to the mathematical proof;
4. assumptions that are *not* made in LF2.

This division is important both mathematically and conceptually. Mathematically, it clarifies what LF2 actually establishes. Conceptually, it prevents the significance of the formal layer from becoming stronger than the theorem package warrants. In the programme dependency map, Layer 2 is already explicitly split into a measure-bridge step and a Born-weight step, with distinct dependency statuses. LF2 should preserve that same separation.

A.2 Internally proved statements

The internal contribution of LF2 is the formal theorem chain that carries the development from the declared ontic-epistemic setup to the projective weight layer. At the level of mathematical content, the following components should be proved directly within the LF2 development.

Pushforward measure construction: Given a measurable map

$$\pi : \Sigma \to CP^{N-1}$$

and a finite ontic measure μ_L on Σ, the formalisation constructs the pushforward measure

$$\pi_* \mu_L$$

on CP^{N-1} and proves the standard rewriting identity

$$(\pi_* \mu_L)(A) = \mu_L(\pi^{-1}(A))$$

for measurable $A \subseteq CP^{N-1}$.

Projective weight construction: Given a preparation measure $\mu_{\text{prep}} \ll \mu_L$ and measurable outcome regions

$$\{\Omega_i^{\text{ep}}(M)\}_{i\in I} \subseteq CP^{N-1},$$

the development defines the projective weights

$$w_i := (\pi_*\mu_{\text{prep}})\big(\Omega_i^{\text{ep}}(M)\big)$$

and proves their basic properties, including non-negativity and normalisation.

Measure-bridge theorem schema: Under the explicitly stated symmetry hypotheses, the formalisation proves that the pushforward measure is invariant under the induced SU(N) action on CP^{N-1}. Given the imported invariant-measure uniqueness theorem, the internal formal step then yields the proportionality statement

$$\pi_*\mu_L = c \cdot \mu_{FS}$$

for some constant c.

Interface theorem with LF1: A key internal lemma identifies the LF1 ontic weight with the LF2 projective weight:

$$\mu_{\text{prep}}\big(\pi^{-1}(\Omega_i^{\text{ep}}(M))\big) = (\pi_*\mu_{\text{prep}})\big(\Omega_i^{\text{ep}}(M)\big).$$

This is the exact formal interface between the deterministic frequency theorem of LF1 and the projective weight layer developed in LF2.

These internal theorems constitute the genuine contribution of LF2 as a formalisation paper. They are not merely inherited from an external library. They are the machine-checked assembly of the theorem spine required for the measure-and-weight layer identified in the programme map. LF1 already established the statistical backbone; LF2 adds the measure bridge and formal weight interface above it.

A.3 Imported mathematical results

LF2 does not attempt to rebuild all required mathematics from first principles. As in LF1, that would be the wrong target. The goal is to formalise a particular layer of the CSD stack, not to rederive the full mathematical substrate of measure theory, homogeneous-space theory, and quantum probability. LF1 already made this explicit at the level of imported Mathlib infrastructure, and LF2 should do the same for its own load-bearing external inputs.

The following ingredients are expected to be imported rather than reproved inside LF2.

Basic definitions and theorems concerning measurable spaces, measurable maps, pushforward measures, absolute continuity, Radon–Nikodym derivatives, and finite measures belong to the imported foundational layer.

The uniqueness of the $SU(N)$-invariant Borel probability measure on CP^{N-1} is treated as an external theorem input. In programme terms, this is the content of the measure-uniqueness step tracked as R9 in CSD1 and developed at the paper level in TN1. LF2 uses that result rather than reproving it from scratch.

A rank-one density uniqueness lemma is also imported in the current LF2 implementation. This lemma states that a density operator assigning certainty to a rank-one projector must itself be the corresponding rank-one density operator. It is used only in the strengthened pure-state endpoint of the Born-weight layer.

The role of imported results in LF2 should be described with care. They are not weaknesses of the formalisation. On the contrary, they are part of what makes the project modular and auditable. What matters is that the paper identifies them explicitly, so that the reader can distinguish the internal LF2 contribution from the broader mathematical infrastructure on which it relies.

A.4 Physical-model assumptions

The most important conceptual benefit of the audit is the isolation of the physical assumptions. These are not proved in Lean. They are not consequences of the measure-theoretic setup alone. They are the modelling inputs required to interpret the formal development as a statement about the finite-dimensional CSD framework.

Existence of an ontic space and Liouville measure: LF2 assumes an ontic space Σ equipped with a physically relevant finite measure μ_L. The paper does not derive this ontology. It states what follows once such a space and measure are given.

Existence of an epistemic projection: LF2 assumes a smooth, surjective, many-to-one map

$$\pi : \Sigma \to CP^{N-1}.$$

The paper does not derive the existence or uniqueness of this projection. It only formalises the consequences of using it in the measure-and-weight layer.

Symmetry-compatible sector assumptions: The measure bridge is conditional on working in the symmetry-compatible quantum-effective sector. Concretely, LF2 assumes $SU(N)$-equivariance of the projection and invariance of μ_L under the corresponding lifted action. In CSD1, this is exactly the sector-conditional content of A4b and R9.

Existence of measurable outcome regions: LF2 assumes that a fixed measurement context determines measurable outcome regions on CP^{N-1} with the required null-boundary regularity. The paper does not classify which such partitions are physically realised. As CSD1 records, that broader universality question remains conditional or open.

Operational consistency package: The Born-weight wrapper depends on the operational consistency package, defined once in Section 5.1 as Definition 5.1. LF2 does not derive these conditions from deeper ontology. It records them as the operational assumptions under which the external theorem inputs apply.

A.5 Assumptions not made

A useful audit should record not only what LF2 assumes, but also what it does *not* assume.

No derivation of Σ or π: LF2 does not claim to derive the ontic manifold Σ or the projection map π from deeper principles.

No fully internal derivation of Born weights: LF2 does not claim a fully internal, self-contained proof of the Born rule. The Born-weight wrapper explicitly depends on the external Busch effect-Gleason theorem and, in its strengthened pure-state endpoint, on an additional imported rank-one density uniqueness lemma.

No resolution of the A5 problem: LF2 does not derive the quantum-effective sector or explain why physical Hamiltonians lie in it. CSD1 treats this as a deferred foundational extension.

No mixed-state, POVM, or sequential-measurement closure: LF2 does not formalise the later finite-dimensional operational dictionary. Those layers belong to LF3 and LF5, corresponding to TN4 and TN6.

No continuum or field-theoretic extension: LF2 remains entirely finite-dimensional. It does not claim any extension to continuum, infinite-dimensional, relativistic, or field-theoretic settings. That discipline is consistent with the current status of the broader programme, where such extensions are deferred to later bridge work.

A.6 Audit summary

The internal contribution of LF2 is the formal measure-and-weight interface: pushforward construction, projective weights, sector-conditional measure bridge, and the explicit connection back to the LF1 ontic weights. The imported mathematical layer consists primarily of invariant-measure uniqueness on CP^{N-1}, the Busch effect-Gleason theorem required for the Born-weight wrapper, and the rank-one density uniqueness input used in the strengthened pure-state endpoint. The physical-model assumptions are the ontic-epistemic architecture and the symmetry-compatible sector conditions. The paper does not claim deeper derivations beyond this layer.

This is exactly the scope LF2 should have. In the current programme map, LF2 is defined as the machine-verified measure bridge and Born-weight wrapper, with the external theorem inputs made explicit as imports rather than hidden inside the narrative. The present audit makes that status precise.

B Internal and External Theorem Boundary

B.1 Purpose

One purpose of LF2 is to make explicit which parts of the measure-and-weight layer are formalised internally and which parts enter as external mathematical inputs. This is important because the present paper sits at a boundary point in the programme. It connects the deterministic frequency theorem of LF1 to the finite-dimensional Born-weight layer, but it does not collapse those two layers into a single theorem.

Accordingly, LF2 distinguishes between:

1. internally formalised results, proved within the LF2 development;
2. external theorem inputs, imported rather than reproved;
3. physical-model assumptions, which belong to the finite-dimensional CSD framework rather than to the internal mathematical proof.

B.2 Internal LF2 theorem layer

The internal LF2 theorem layer consists of the formal measure-theoretic bridge from ontic weights to projective weights.

More precisely, LF2 proves the following schematic chain.

Pushforward construction: Given a measurable projection

$$\pi : \Sigma \to CP^{N-1}$$

and a finite ontic measure μ_L on Σ, LF2 defines the pushforward measure

$$\pi_*\mu_L$$

and proves the standard identity

$$(\pi_*\mu_L)(A) = \mu_L(\pi^{-1}(A))$$

for measurable $A \subseteq CP^{N-1}$.

Invariance transfer: Given the explicit symmetry hypotheses of the symmetry-compatible sector, LF2 proves that $\pi_*\mu_L$ is invariant under the induced $SU(N)$ action on CP^{N-1}.

Projective weight construction: Given a preparation measure μ_{prep} on Σ, LF2 defines the corresponding projective weights

$$w_i = (\pi_*\mu_{\text{prep}})\big(\Omega_i^{\text{ep}}(M)\big)$$

for measurable outcome regions $\Omega_i^{\text{ep}}(M) \subseteq CP^{N-1}$, and proves their basic measure-theoretic properties, including non-negativity and normalisation.

Interface with LF1: LF2 proves that the LF1 ontic weight and the LF2 projective weight coincide:

$$\mu_{\text{prep}}\big(\pi^{-1}(\Omega_i^{\text{ep}}(M))\big) = (\pi_*\mu_{\text{prep}})\big(\Omega_i^{\text{ep}}(M)\big).$$

This identity is the exact formal interface between the deterministic repeated-trial theorem of LF1 and the projective weight layer of LF2.

B.3 External mathematical inputs

LF2 does not reprove all mathematical results needed for the full finite-dimensional probability layer. The following inputs are treated as external.

Invariant-measure uniqueness on CP^{N-1}: LF2 imports the uniqueness of the SU(N)-invariant Borel probability measure on CP^{N-1}. From this result, together with the internally proved invariance of $\pi_*\mu_L$, LF2 obtains the sector-conditional proportionality

$$\pi_*\mu_L = c \cdot \mu_{FS}$$

for some constant c, Busch effect-Gleason theorem for N $\geq$ 2: LF2 imports the finite-dimensional theorem that, under the effect-additive operational consistency package of Definition 5.1, the admissible probability assignment must take trace form for all finite dimensions N $\geq$ 2.

These results are not hidden assumptions. They are explicit imported inputs, and LF2 is designed to record them as such.

B.4 Physical-model assumptions

In addition to external theorem inputs, LF2 depends on physical-model assumptions belonging to the finite-dimensional framework.

Ontic space and Liouville measure: LF2 assumes an ontic space Σ equipped with a finite reference measure μ_L.

Epistemic projection: LF2 assumes a smooth, surjective, many-to-one map

$$\pi : \Sigma \to CP^{N-1}.$$

Symmetry-compatible sector: LF2 assumes the symmetry hypotheses under which the measure bridge is expected to hold, including $SU(N)$-equivariance of π and invariance of μ_L under the corresponding lifted action.

Outcome-region structure: LF2 assumes that each measurement context determines measurable projective outcome regions with the required null-boundary regularity.

These assumptions are part of the model architecture. They are not proved internally in LF2.

B.5 What LF2 does not claim

The explicit theorem boundary also clarifies what LF2 does not claim.

LF2 does not claim:

- a derivation of the ontic manifold Σ;
- a derivation of the projection map π;
- a fully internal proof of invariant-measure uniqueness on CP^{N-1};
- a fully internal proof of the Busch effect-Gleason theorem;
- a fully internal proof of the rank-one density uniqueness lemma used in the pure-state endpoint;
- a reconstruction of the full finite-dimensional operational dictionary beyond the measure-and-weight layer.

In particular, LF2 should not be read as a standalone first-principles derivation of the whole finite-dimensional quantum formalism. Its role is narrower: it formalises the measure bridge and packages the Born-weight wrapper under clearly stated external inputs.

B.6 Summary

The internal content of LF2 is the formal measure-and-weight interface:

$$\text{ontic measure} \;\longrightarrow\; \text{pushforward measure} \;\longrightarrow\; \text{projective weights}.$$

The external content consists of the invariant-measure uniqueness theorem and the Busch effect-Gleason theorem and the rank-one density uniqueness input needed for the implemented pure-state Born-weight layer. The physical-model content is the ontic-epistemic architecture within which these theorems are applied.

This boundary is not a weakness of LF2. It is part of its purpose. The paper is intended to make explicit exactly which parts of the finite-dimensional probability layer are internally formalised, which are imported, and which are assumed as part of the underlying model.

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